CHAPTER

32

The Laplace Transform

The two main techniques in signal processing, convolution and Fourier analysis, teach that a linear system can be completely understood from its impulse or frequency response. This is a very generalized approach, since the impulse and frequency responses can be of nearly any shape or form. In fact, it is *too* general for many applications in science and engineering. Many of the parameters in our universe interact through *differential equations*. For example, the voltage across an inductor is proportional to the derivative of the current through the device. Likewise, the force applied to a mass is proportional to the derivative of its velocity. Physics is filled with these kinds of relations. The frequency and impulse responses of these systems cannot be arbitrary, but must be consistent with the solution of these differential equations. This means that their impulse responses can only consist of *exponentials* and *sinusoids*. The Laplace transform is a technique for analyzing these special systems when the signals are *continuous*. The z-transform is a similar technique used in the *discrete* case.

The Nature of the s-Domain

The Laplace transform is a well established mathematical technique for solving differential equations. It is named in honor of the great French mathematician, Pierre Simon De Laplace (1749-1827). Like all transforms, the Laplace transform changes one signal into another according to some fixed set of rules or equations. As illustrated in Fig. 32-1, the Laplace transform changes a signal in the time domain into a signal in the **s-domain**, also called the **s-plane**. The time domain signal is continuous, extends to both positive and negative infinity, and may be either periodic or aperiodic. The Laplace transform allows the time domain to be *complex*; however, this is seldom needed in signal processing. In this discussion, and nearly all practical applications, the time domain signal is completely real.

As shown in Fig. 32-1, the s-domain is a complex plane, i.e., there are real numbers along the horizontal axis and imaginary numbers along the vertical axis. The distance along the real axis is expressed by the variable, σ, a lower